Taylor's Series

Taylor's Polynomial:-

f is a function

$\downarrow$ approximate

P(x) a polynomial

$$P(x) = a_0 + a_1 x + \cdots + a_n x^n$$

$$f(x) = P(x) + \text{Error term}$$

Taylor Polynomial:-

Let f be a function having an n^{th} derivative at a point x_0

The polynomial $P_n(x)$ defined as

$$P_n(x) = f(x_0) + (x-x_0) f'(x_0) + (x-x_0)^2 \frac{f''(x_0)}{2!} + \cdots (x-x_0)^n \frac{f^n(x_0)}{n!}$$

is called the n^{th} degree Taylor's Polynomial for f at x_0.

→ Polynomial which is used to approximate a function is called Taylor Polynomial

$$P_n(x) = f(x_0) + (x-x_0)\frac{f'(x_0)}{1!} + (x-x_0)^2\frac{f''(x_0)}{2!} + \dots + (x-x_0)^n\frac{f^n(x_0)}{n!} \quad \text{— } (*)$$

$$P_n(x_0) = f(x_0) + \underbrace{(x_0-x_0)}_{0}\frac{f'(x_0)}{1!} + \underbrace{(x_0-x_0)^2}_{0}\frac{f''(x_0)}{2!} + \dots + (x_0-x_0)^n\frac{f^n(x_0)}{n!}$$

$$P_n(x_0) = f(x_0)$$

Differentiate $(*)$ w.r.t x, we have

$$P_n'(x) = f'(x_0) + \cancel{2}(x-x_0)\frac{f''(x_0)}{\cancel{2!}} + \dots + \cancel{n}(x-x_0)^{n-1}\frac{f^n(x_0)}{\cancel{n!}\,(n-1)!}$$

$$= f'(x_0) + (x-x_0)f''(x_0) + \dots + (x-x_0)^{n-1}\frac{f^n(x_0)}{(n-1)!}$$

$$P_n'(x_0) = f'(x_0)$$

$\vdots$

$$P_n^{n}(x_0) = f^{n}(x_0) \quad \forall \; 1 \le k \le n$$

At point $x = x_0$

$$P_n(x_0) = f(x_0)$$

and also $P_n'(x_0) = f'(x_0)$

i.e. Slope of tangent of f at x_0 equal to slope of tangent of Polynomial P_n at point x_0.

$\rightarrow$ The Taylor's Polynomial for 'f' at 'x_0' is a Polynomial which approximates 'f' for points near x_0

Eg:- Find the 4^{th} degree Taylor's Polynomial for $f(x) = \cos x$ at $x_0 = 0$

&

$$P_4(x) = f(x_0) + (x - x_0)\frac{f'(x_0)}{1!} + (x - x_0)^2 \frac{f''(x_0)}{2!}$$

$$+ \frac{(x-x_0)^3}{3!} f'''(x_0) + \frac{(x-x_0)^4}{4!} f^{iv}(x_0)$$

$f(x) = \cos x$ $\qquad$ $f(0) = 1$

$f'(x) = -\sin x$ $\qquad$ $f'(0) = 0$

$f''(x) = -\cos x$ $\qquad$ $f''(0) = -1$

$f'''(x) = \sin x$ $\qquad$ $f'''(0) = 0$

$f^{iv}(x) = \cos x$ $\qquad$ $f^{iv}(0) = 1$

$$P_4(x) = f(0) + (x-0)\frac{f'(0)}{1!} + (x-0)^2\frac{f''(0)}{2!}$$

$$+ (x-0)^3\frac{f'''(0)}{3!} + (x-0)^4\frac{f^{iv}(0)}{4!}$$

$$= 1 + 0 + x^2\frac{(-1)}{2!} + 0 + x^4\frac{(1)}{4!}$$

$$\therefore P_4(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

Questions

1) Find the 3rd degree Taylor's Polynomial for $f(x) = e^x$ at $x_0 = 0$

2) Find the 3rd degree Taylor's Polynomial for $f(x) = (1+x)^{1/3}$ at $x_0 = 0$

Taylor's Theorem:- (How the Taylor's Polynomial approximate the function)

Let $n \in \mathbb{N}$, $I = [a, b]$ and $f: I \to \mathbb{R}$ such that

1) $f', f'', \dots, f^{n}$ are continuous on $I = [a, b]$
2) f^{n+1} exists on (a, b)

If $x_0 \in I$, then for any $x \in I$, $\exists$ c in between x and x_0 such that

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0)\frac{(x-x_0)^2}{2!}$$

$$+ \dots + f^{n}(x_0)\frac{(x-x_0)^{n}}{n!} + f^{n+1}(x_0)\frac{(x-x_0)^{n+1}}{(n+1)!}$$

$$f(x) = P_n(x) + R_n(x)$$

when $\langle R_n(x) \rangle \to 0$ then $P_n(x) \to f(x)$

Then $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{(x-x_0)^{n}}{n!} \quad \forall x$

Such that $\langle R_n(x) \rangle \to 0$

$$\boxed{\text{And } f(x) = \sum_{n=0}^{\infty} f^{n}(x)\,\frac{(x-x_0)^n}{n!} \text{ is Taylor's series for the function } f \text{ at } x = x_0}$$

Expansion of functions

1. Maclaurin's series
2. Expansion by use of known series
3. Taylor's series

Rolle's Theorem and Mean Value Theorem

Rolle's Theorem:-

Let f be a function defined on $[a,b]$ such that
1. f is continuous on $[a,b]$
2. f is differentiable on (a,b)
3. $f(a) = f(b)$

Then $\exists$ a real number $c \in (a,b)$ such that $f'(c) = 0$

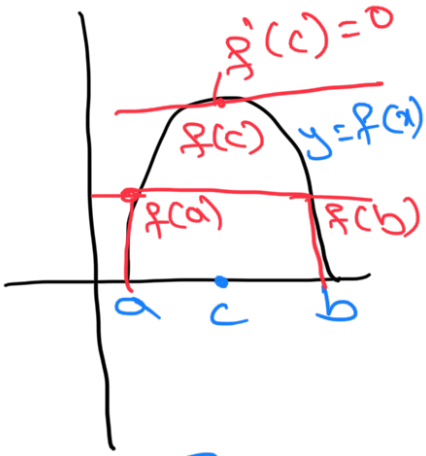


Fig 1

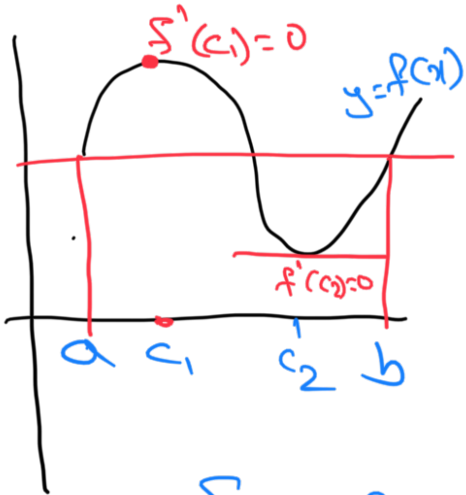


Fig 2

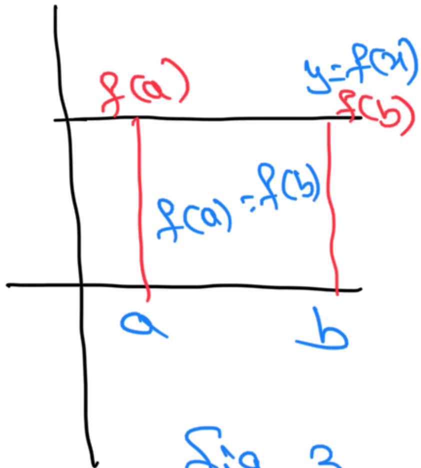


Fig 3

The conclusion of Rolle's Theorem may not hold if any of these condition fails.

(1) $f'(x)$ doesn't exist in (a,b)

eg: $f(x) = |x|$ in $[-1,1]$

✓ (1) $f(x)$ is continuous on $[-1,1]$

✗ (2) $f(x)$ is not differentiable at $x=0$

(3) $f(1) = 1 = f(-1)$

$f'(x) \neq 0 \ \forall x \in (-1,1) - \{0\}$

there doesnot exist any $c \in (-1,1)$ such that $f'(c) = 0$

(2) $f(a) \neq f(b)$

eg $f(x) = x$ in $[1,2]$

but $f(1) = 1$

$f(2) = 2$

$\Rightarrow f(1) \neq f(2)$

$f'(x) = 1 \quad \forall\ x \in (1, 2)$

there doesnot exist any $c \in (1, 2)$ such that $f'(c) = 0$

Proof of Rolle's theorem :-

Pf: we know that a function which is continuous on a closed interval is bounded there in and attains its bound

$\therefore f$ is bounded on $[a, b]$

Let $m \leq f(x) \leq M \quad \forall\ x \in [a, b]$ — (1)

Since f attains its bounds, there exists points d and c in $[a, b]$ such that

$f(d) = m$ and $f(c) = M$ — (2)

Using (1) and (2), we get

$$f(d) \leq f(x) \leq f(c) \quad \forall x \in [a, b] \quad \text{— (3)}$$

Case 1 :- $M = m$

from (1)

$$m \leq f(x) \leq M \;\forall\; x \in [a,b]$$

$$\Rightarrow \quad f(x) = m \;\forall\; x \in [a,b]$$

$$\Rightarrow \quad f'(x) = 0 \;\forall\; x \in [a,b]$$

Thus, the theorem is true in this case

Case 2 :- $M \neq m$

Since, $f(a) = f(b)$ and $M \neq m$

so atleast one of the numbers M and m i.e different from $f(a)$ and $f(b)$

Without loss of generality,

$M \neq f(a)$ and $M \neq f(b)$

$f(c) \neq f(a)$ and $f(c) \neq f(b)$ (using 2)

$\Rightarrow \quad c \neq a$ and $c \neq b$ where $c \in [a,b]$

Thus $c \in (a,b)$ and by condition 2

$f'(c)$ exists

$$f'(c) = Lf'(c) = Rf'(c)$$

Now $Lf'(c) = \lim_{x \to c^-} \dfrac{f(x) - f(c)}{x - c}$

from 3

$f(x) \leq f(c) \Rightarrow f(x) - f(c) \leq 0$

$x \to c^- \Rightarrow x < c \Rightarrow x - c < 0$

$$L f'(c) \geq 0 \quad \text{—— } (*)$$

Now $R f'(c) = \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$

$$x \to c^+ \Rightarrow x > c \Rightarrow x - c > 0$$

$$R f'(c) \leq 0 \quad \text{—— } (**)$$

from $(*)$ and $(**)$

$$f'(c) = 0$$

where $c \in (a, b)$

Hence Proved

Mean Value Theorem

Let f be a function defined on $[a, b]$, such that

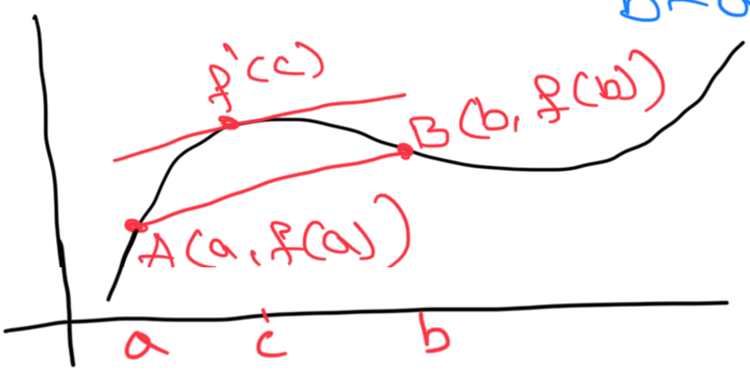
(1) $f(x)$ is continuous on $[a, b]$

(2) $f(x)$ is differentiable on (a, b)

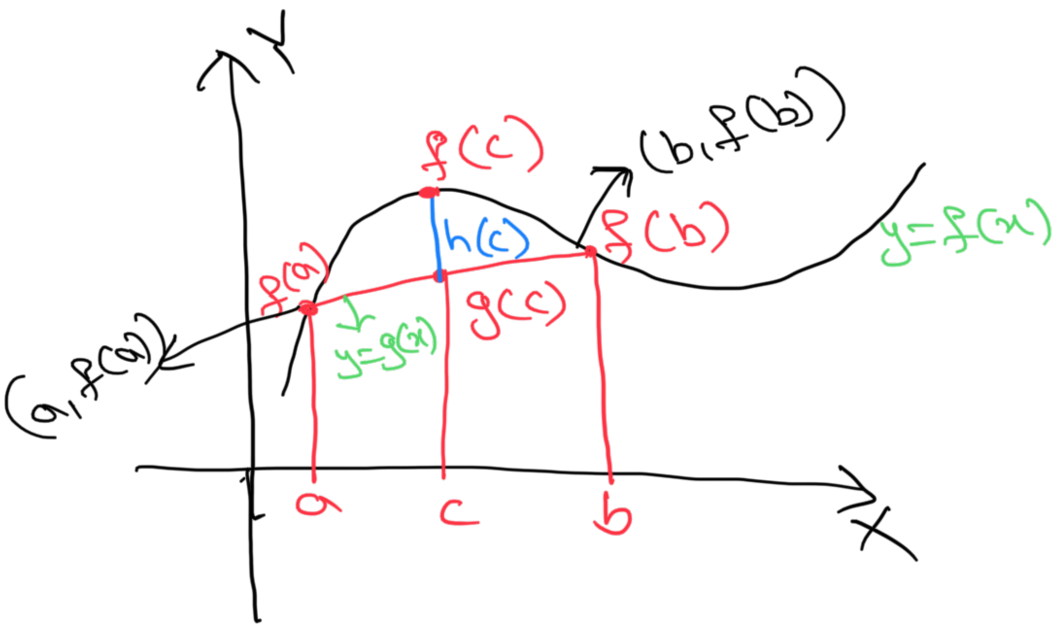
Then there exist some $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

Proof:-



$\exists$ a point corresponding to $c \in (a,b)$ on the curve $y = f(x)$ at which the tangent to the curve is parallel to the chord.



Equation of line joining AB is given by

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y-f(a)}{x-a} = \frac{f(b)-f(a)}{b-a}$$

Let $g(x) = y = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$

Define $h(x) = f(x) - g(x)$

$= f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$ — 1

Let us calculate the value of the function at end points of interval i.e a and b

$h(a) = 0$

$h(b) = f(b) - f(a) - \frac{f(b)-f(a)}{b-a}(b-a)$

$= 0$

$\Rightarrow h(a) = h(b)$

$\therefore$ we have (1) $h(x)$ is continuous on $[a, b]$

(2) $h(x)$ is differentiable in (a, b)

(3) $h(a) = h(b)$

All the conditions of Rolle's theorem are satisfied.

There exists point $c \in (a, b)$ such that $h'(c) = 0$ — 2 from ①

$$h'(x) = f'(x) - 0 - \left[\frac{f(b)-f(a)}{b-a}\right](1-0)$$

$$h'(x) = f'(x) - \left[\frac{f(b)-f(a)}{b-a}\right]$$

from ②, we have

$$h'(c) = 0$$

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

Hence Proved

1) Verify Rolle's theorem in $[-1, 1]$ for the function $f(x) = x^2$

Sol

$$f(a) = f(b)$$

$$f(-1) = f(1)$$

$$(-1)^2 = (1)^2$$

$$\exists\ c \in (-1, 1)$$

$$f'(c) = 0$$

$$2c = 0$$

$$c = 0 \in (-1, 1)$$

$\therefore$ Rolle's theorem verified

2) verify Rolle's theorem in $[2, 4]$ for the function $f(x) = x^2 - 6x + 8$

Sol

$$f(2) = 2^2 - 6 \times 2 + 8 = 0$$
$$f(4) = 4^2 - 6 \times 4 + 8 = 0$$

$$f(2) = f(4)$$

$\exists\ c \in (2, 4)$ s.t $f'(c) = 0$

where $c \in 3 \in (2, 4)$

$\therefore$ Rolle's theorem verified

Problems:-

1) Verify Rolle's theorem for

$f(x) = \sqrt{1 - x^2}$ in $[-1, 1]$

2) $f(x) = x(x+3)e^{-\frac{1}{2}x}$ in $[-3, 0]$